2412ET107

Roll No. 2349bc0910103

Course: B.Tech Computer Engineering / B.Tech COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Subject: Signal & Systems

Subject Code: ETCS-211

Semester: III

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q. No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a	Define the following Signals mathematically: i. Even and Odd Signals ii. Causal and Non-Causal Signals	(2)
b	Distinguish between Static and dynamic systems with suitable example.	(2)
c	What are Dirichlet's conditions? State them.	(2)
d	Does Fourier transform exist for functions which are not absolutely integrable?	(2)
e	What are the limitation of Fourier series? Does Fourier transform overcome this?	(2)
f	How is Laplace transform **useful** in the analysis of LTI systems?	(2)
g	What are the advantages and limitations of Z-transform?	(2)
h	State the linearity property of Z-transform.	(2)
i	Define inverse discrete-time Fourier transform.	(2)
j	What is the relationship between discrete-time Fourier transform and Z-transform?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. a. Derive the relationship between complex exponential and sinusoidal signals. (10)
   b. What is the system? How are systems classified? Define each one of them.
3. With regard to Fourier series representation, justify the following statements: (10)
   (a) Odd functions have only sine terms.
   (b) Even functions have no sine terms.
4. Find the Fourier transform of the signal $x(t) = e^{-a|t|}sgn(t)$, where sgn(t) is sign of t. (10)
5. Using the properties of Laplace transform, find the Laplace transform of the following signals: (10)
   i. $x(t) = te^{-t}u(t)$
   ii. $x(t) = e^{-5t}sin10at\ u(t)$
6. State and prove the time shifting, time reversal and time expansion properties of Z-Transform. (10)
7. Find the Z-transform of (10)
   i. $y(n) = x(n-1)\,u(n)$ ii. $n\delta(n-4)$
8. Realize the following transfer system in direct form-II (10)
$$H(s) = \frac{s+4}{s^2+3s+5}$$
9. State and prove the time reversal and differentiation in frequency domain properties of DTFT. (10)

===END OF PAPER===